

Vatsal Omar

Customer Engagement Platform Engineer

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Customer engagement platform engineer with expertise in building real-time notification systems, multi-channel messaging workflows, and AI-driven personalization. Designed agentic notification pipelines for enterprise clients and developed cross-platform apps with sub-second latency for mission-critical alerts.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

2023-08 – 2026-07

Bachelor of Science in Computer Science — Pilani, IN

EXPERIENCE

Software Development Intern

2024-12 – 2025-09

BLive — Bangalore, IN

- Architected agentic notification workflow using Firebase Cloud Messaging and Python backend to route mission-critical alerts (SOS, delivery updates) to operator roles with sub-1000ms latency
- Built real-time telematics intelligence layer processing EV IoT data streams with rule-based and ML-assisted anomaly detection for battery health and route deviation alerts
- Integrated Google Maps with live geofencing and route optimization for operations teams, reducing fleet monitoring latency by 30%
- Developed two cross-platform Flutter apps (rider delivery + fleet dashboard) with shared Dart codebase, achieving 25% faster load times via rendering optimization

PROJECTS

StrategyVault — AI Trading Strategy Marketplace

FastAPI · Next.js 14 · Redis · PostgreSQL

Subscription-based platform using multi-model consensus engine (Gemini 2.0, OpenAI, Claude) to validate and rank institutional-grade trading strategies

- Built parallel multi-model consensus engine producing BUY/HOLD/REJECT verdicts with confidence rankings via Gemini 2.0, OpenAI, and Claude integration
- Engineered RBG (Research-Backtest-Implement) agent loop with structured prompt refinement for autonomous strategy generation and validation

AcordLayer — Self-Hosted Collaboration Platform

Go · TypeScript · React · PostgreSQL

Privacy-focused collaboration platform with real-time messaging and workflow automation built with Go and React

- Designed WebSocket-based real-time messaging system with threaded channels for low-latency multi-user collaboration
- Implemented no-code workflow automation layer with containerized task execution for agentic workflow orchestration

ACHIEVEMENTS

Top 8 — OpenEnv AI Hackathon — Ranked top 8 out of 70,000+ developers in India's largest AI hackathon (Meta × Hugging Face × PyTorch), building RL environments on Meta's OpenEnv

Intern of the Quarter — BLive — Recognized for technical excellence in building real-time notification systems and cross-platform apps for enterprise clients

TECHNICAL SKILLS

Languages: TypeScript · Python · Dart · JavaScript · SQL

Frontend: Flutter · React · Next.js · Tailwind CSS

Backend: FastAPI · NestJS · Firebase · PostgreSQL · Redis

Ai ML: Multi-model LLM integration · Prompt Engineering · Function-calling · OpenAI GPT · Anthropic Claude · Google Gemini

Cloud Infra: Google Cloud Platform · Docker · CI/CD · REST APIs